
1. Experiment: Rosetta Evaluation and MD Simulation for TOP Candidate Selection**2. Time:** 2026.02.01-2026.02.7**3. Member:** Kaiqing Zhang, Yang Jin, Binxuan Zhang, Qiwen Jiang, Yue Fu, Ziyang Yu**4. Method:**

(1) Rosetta Static Interface Evaluation

Following the first round of screening and sequencing, three combinations were identified as the top enriched variants: A2B3C3, A2B3C2, and A3B2C2. To evaluate their binding interfaces at the structural level, we assessed these three candidates alongside the gp130 reference complex using Rosetta static interface metrics, including interface energy (REU), buried interface area, energy density (interface energy per unit area), and interface packing score, before subjecting them to molecular dynamics simulations.

(2) System Preparation

- ① We used the Rosetta-refined structures of the three candidates (A2B3C3, A2B3C2, A3B2C2) and the gp130 reference complex in complex with IL-6/sIL-6R α as the starting conformations for MD simulations. The gp130 reference complex was derived from the IL-6/sIL-6R α /gp130 ternary crystal structure (PDB: 1P9M).
- ② Each complex was placed in a periodic cubic water box with a minimum protein-to-box distance of 12 Å. The systems were solved with the TIP3P water model, neutralized with Na⁺ and Cl⁻ counterions, and adjusted to a physiological salt environment. Proteins were parameterized with the CHARMM36m force field; water and ion parameters were taken from the corresponding CHARMM-compatible parameter set.

(3) Molecular Dynamics Simulation Settings

- ① All molecular dynamics simulations were performed using GROMACS 2022.5. Each solvated system underwent steepest-descent energy minimization, followed by sequential NVT and NPT equilibration, and a 100 ns production simulation.
- ② During NVT equilibration, the system was equilibrated for 1 ns with positional restraints applied to the protein backbone to permit relaxation of side chains and solvent molecules. This was followed by NPT equilibration using Berendsen pressure coupling, while positional restraints on the solute were gradually reduced.
- ③ The 100 ns production simulation was conducted in the NPT ensemble. Temperature was maintained with a velocity-rescaling thermostat, and pressure was controlled using the Parrinello-Rahman coupling method.
- ④ Short-range van der Waals and electrostatic interactions were evaluated using the prescribed cutoff scheme, whereas long-range electrostatics were treated with the particle mesh Ewald (PME) method.

(4) Data Analysis

- ① Before trajectory analysis, periodic boundary artifacts were removed, each protein complex was centered, and the trajectories were fitted to eliminate overall translational and rotational motions.
- ② We calculated C α RMSD separately for the receptor (IL-6/sIL-6R α) and for each designed binder (or gp130) to track their individual conformational stability. We counted interfacial hydrogen bonds using a donor-acceptor distance cutoff of 3.5 Å, and calculated the buried solvent-accessible surface area (BSA) with the gmx sasa tool. We also computed per-residue RMSF/B-factor, and extracted van der Waals and electrostatic interaction energies along the trajectory. These quantities were summarized in comparative radar charts for cross-evaluation of the four complexes.
- ③ We constructed relative free-energy landscapes from the first two principal components (PC1 and PC2) derived from the C α covariance matrix, with $\Delta G = -k_B T \ln[P(PC_1, PC_2)/P_{\max}]$. We used VMD to visualize trajectories and extracted representative snapshots from t = 20 ns (initial, after equilibration)

and t = 100 ns (final) to compare binding-mode changes.

- ④ The initial 20 ns of each 100 ns production trajectory was treated as an additional structural-relaxation period and excluded from the primary statistical analysis. Mean values and standard deviations were calculated over the remaining 80 ns, and the final 20 ns was separately examined for late-stage conformational drift or interface rearrangement.

5. Result:

Table 1. PIA-derived interface characteristics of the analyzed protein-protein complexes.

System	Interfacial Residue Pairs	Hydrogen Bonds (HBs)	Salt Bridges	van der Waals (vdW) Clashes	Mean Positive Shape Complementarity (SC)	Mean Ligand Residue Burial (%)
candidate1 (A2B3C3)	68	17	5	0	0.628	51.4
candidate3 (A3B2C2)	43	3	2	0	0.701	60.1
candidate2 (A2B3C2)	68	13	5	0	0.574	54.4
Reference (GP130)	102	9	2	2	0.715	56.1
candidate1- sIL-6Rα	72	8	3	0	0.644	59.2
candidate1- IL-6	53	17	2	0	0.608	58.4

Note. Interface statistics were calculated from one representative structure for each system. Interfacial residue pairs denote cross-interface residue-residue contacts identified by PIA. HBs, salt bridges, and van der Waals (vdW) clashes indicate the corresponding interaction counts. Mean positive shape complementarity (SC) and mean ligand-residue burial were reported as calculated by PIA. The candidate1-sIL-6Rα and candidate1-IL-6 complexes were generated by independent docking and were not obtained by deleting one chain from the complete candidate 1 complex. These static structural descriptors should not be interpreted as binding free energies or molecular-dynamics trajectory averages.

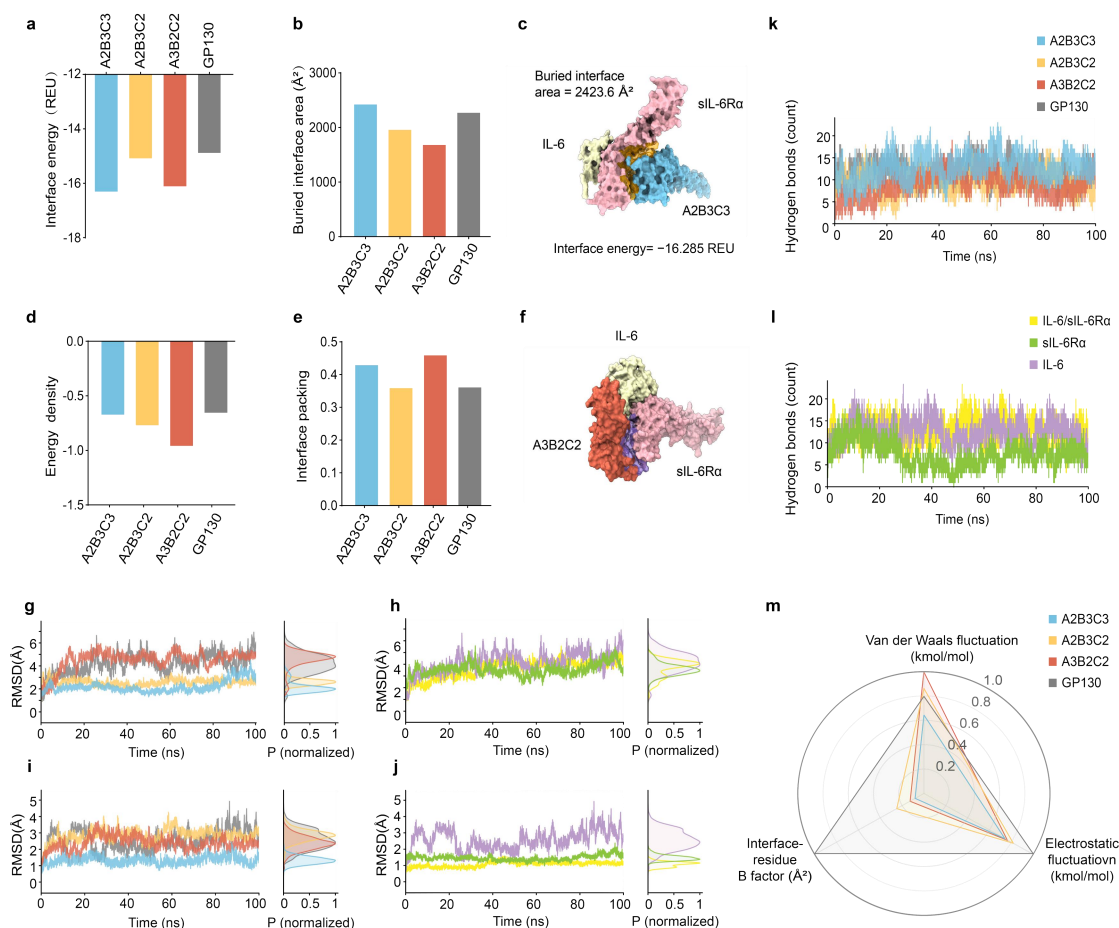


Figure 1. Overview of the four systems subjected to Rosetta evaluation and 100 ns MD simulations. a, b, Rosetta-derived interface energy (a) and buried interface area (b) of representative optimized models of A2B3C3, A2B3C2 and A3B2C2, with native gp130 included as the reference. c, Structural model of A2B3C3 bound to the IL-6/sIL-6R α complex, showing its buried interface area and Rosetta interface energy. d, e, Interface energy density (d) and interface packing score (e) of the three candidate binders and gp130. f, Structural model of A3B2C2 bound to the IL-6/sIL-6R α complex, illustrating its compact binding interface. g-j, Root-mean-square deviation (RMSD) trajectories and corresponding normalized probability distributions obtained from 100-ns molecular-dynamics simulations. Candidate-protein or gp130 RMSDs are shown in g, and IL-6/sIL-6R α receptor-side RMSDs are shown in i; component-resolved RMSD analyses of the A2B3C3-bound system are shown in h and j. k, Number of interfacial hydrogen bonds formed during the 100-ns simulations of the three candidate complexes and the gp130 reference complex. l, Component-resolved hydrogen-bond analysis of A2B3C3 with the complete IL-6/sIL-6R α complex or with the individual IL-6 and sIL-6R α components. m, Radar plot comparing the normalized fluctuations in interfacial van der Waals energy, electrostatic energy and interface-residue B factors.

6. AI Use Statement:

Computational prediction generated using AI, not experimentally validated by wet experiment. We employed a combined approach of computational design and experimental verification. Therefore, for results from AI-assisted prediction/simulation, we will continue to conduct wet experimental verification to confirm their validity.